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PALAEOMAGNETIC STUDY OF THE SWEDISH RAPAKIVI SUITE: PROTEROZOIC TECTONICS OF THE BALTIC SHIELD

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The major Proterozoic igneous intrusions in the Swedish sector of the Baltic Shield are the Ragunda complex (1293 m.y., palaeomagnetic pole 165°E, 54°N) and the Nordingrå gabbro-granite-anorthosite complex (1385 ± 30 m.y.). The latter body has been partially remagnetised by later post-Jotnian dolerites (1254 m.y.), and sites influenced by the dolerites have a stable magnetisation with a mean direction $D = 45°$, $I = -39°$, ($\alpha_{95} = 4.3°$). Elsewhere, the gabbro-anorthosite facies have a magnetisation of dual polarity predating the dolerite and recoverable at various stages of thermal and/or a.f. cleaning with a mean of $D = 48°$, $I = 37°$ ($\alpha_{95} = 5.3°$); medium and high coercivity remanence resides in large magnetite grains and fine, predominantly hematite, rods in feldspar megacrysts. The Nordingrå rapakivi granite yields a mean, also including dual polarities, of $D = 221°$, $I = -25°$ ($\alpha_{95} = 13°$), and the Gävle granite yields a mean of $D = 26°$, $I = 17°$ ($\alpha_{95} = 13°$).

New data define the a.p.w. path for the Baltic Shield after final uplift and cooling of the ca. 1800 m.y. Svecofennian mobile belt and prior to intrusion of the post-Jotnian dolerites at 1250 m.y.; this (ca. 1500–1200 m.y.) path defines a double loop similar in size and shape to the contemporaneous path for the Laurentian Shield and the paths can be superimposed to define relative positions of the shields. They were in juxtaposition prior to 1200 m.y. with the optimum reconstruction obtained by rotation of approximately 64° about a Euler pole at 1°E, 36°N. Pre-1500 m.y. palaeomagnetic data are also shown to fit this same unique reconstruction. The main geological correlations are an alignment of the Lower/Middle Proterozoic major strike-slip zones, the structural trends within the pre-1700 m.y. mobile belts, and the Grenville and Sveconorwegian (ca. 1100 m.y.) mobile belts. The anorogenic magmatism characteristic of Proterozoic times became gradually more restricted to one active margin of the continental reconstruction as temperature gradients decreased and the crust consolidated. All of these Proterozoic tectonic/magmatic trends are parallel to the long axis of the continental reconstruction.

1. Introduction

Stabilisation and uplift of the widespread ca. 1800 m.y. (Churchill-Hudsonian, Ketilidian, Svecofennian-Svecokarelian) mobile belts within the Laurentian, Baltic and Ukrainian Shields, were followed by the intrusion of distinctive igneous rock suites in Middle-Upper Proterozoic times [1]. These include large plutons of massive anorthosites, rapakivi granites and alkaline complexes, basaltic and lamprophyric dyke swarms, and extensive suites of plateau basalts and ignimbrites; this was apparently the first time in Earth history that such associations began to form prominent rock groups [2]. Within these shields, they provide the main opportunity to sample the palaeomagnetic field between the time of cooling of the ca. 1800 m.y. mobile belts and development of the late Proterozoic Sveconorwegian and Grenville mobile belts at ca. 1100–1000 m.y.

This paper reports a palaeomagnetic study of rapakivi-granite/massive anorthosite bodies within the Swedish part of the Baltic Shield. It also discusses the definition of the Proterozoic apparent polar wander (a.p.w.) path for this shield and examines the correlation between this path and the contemporaneous data from the Laurentian Shield. The geological implications of these correlations are briefly examined.

2. Geological setting

The bulk of the Swedish sector of the Baltic Shield consists of gneisses and granites affected by the Svecofennian (or Svecokarelian) mobile episode ending about 1800 m.y. [3]. The termination of tectonic activity is defined by the Rb-Sr isochron age of 1748 ± 40 m.y. * [4] on the late- to post-tectonic Revsund granite. The intrusion of post-tectonic granites within the belt was very widespread [5], and it is likely that uplift and cooling took place over a time interval of up to several hundreds of millions of years after 1750 m.y. [3,5,6].

During and immediately following uplift and cooling of this mobile belt, anorogenic intrusions and deposition of supracrustal successions took place prior to widespread fracturing and intrusion of the terrain by dolerites (the post-Jotnian dolerites at ca. 1270–1210 m.y. [7]). The mid-Proterozoic episodes are predominantly sedimentary/volcanic (Jotnian) in the west but include the major intrusions at Ragunda [8] and Nordingrå [9] in the east (Fig. 1). The Ragunda gabbro-granite complex has been dated by the Rb-Sr method at 1293 ± 20 m.y. (results quoted in Kornfelt [8], and palaeomagnetic data given in Piper [10]). The granitic facies of the Nordingrå granite-gabbro-anorthosite complex is dated 1383 ± 30 m.y. by the Rb-Sr method [11]; both this complex and the Gävle granite intrusion some 250 km to the south ([9] and Fig. 1) are cut by the ca. 1250 m.y. post-Jotnian dolerites.

At Nordingrå the gabbro-anorthosite is an earlier phase than the granite and is therefore somewhat older than 1385 m.y.; both are overlain unconformably by a thin suite of sediments lying on a strongly eroded surface and assigned to the Jotnian series [12]. The term "Jotnian" was introduced by Sederholm [13] to embrace predominantly arenaceous post-Svecofennian and pre-dolerite sediments, but there is no definitive evidence to equate the sediments at Nordingrå with the major outcrop of Jotnian sediments in the Dalarna area of western Sweden (Fig. 1). These sediments and the complex are in turn cut by a thick dolerite intrusion ([14] and Fig. 2); this forms part of a large sheet-like body of lopolithic form with

* All ages quoted in this paper have been recalculated to conform with the decay constants listed by Steiger and Jäger [70].

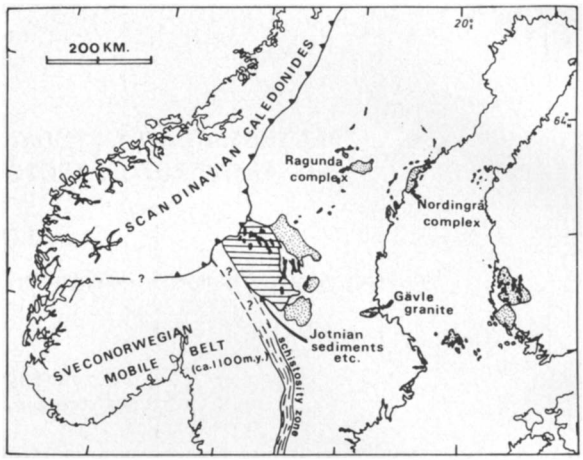


Fig. 1. Distribution of Middle-Upper Proterozoic anorogenic intrusions (stippled), sedimentary-volcanic suites (shaded) and post-Jotnian dolerites (ca. 1200 m.y.) in the western part of the Baltic Shield.

a centre near the western margin of the Nordingrå complex and a thickness of 200–500 m, locally reaching 800 m [15]. The gabbro [14,16] is composed predominantly of labradorite feldspar, hornblende, mica and ores, is saturated or slightly over saturated, and passes gradationally first into coarse-grained labradorite-bearing varieties and then into true anorthosites; gradations to monzonitic and syenitic composition are also observed near to the contacts with the granite. The latter is a typical red rapakivi granite and mafic constituents are in general very subordinate. The outcrop map of Fig. 2 is based on the work of Sobral [12].

3. Palaeomagnetic results

The palaeomagnetic samples were drilled as short cores using a portable two-stroke motor and oriented by solar compass or magnetic compass in conjunction with topographic sights. A sampling site is defined as one drilling locality with samples distributed over several metres of outcrop; the location of these sites is given in Fig. 2.

In the laboratory samples were measured with a parastatic magnetometer system before and after progressive cleaning treatment in a.f. fields incremented in steps of 5 or 10 milli-tesla (mT) to

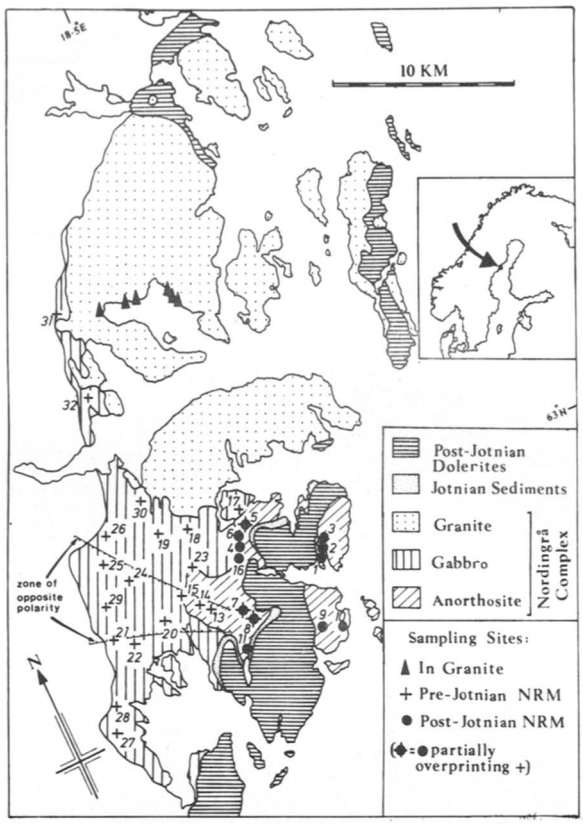


Fig. 2. Geology and sampling sites in the Nordingrå gabbro-granite-anorthosite complex, east Sweden; the inset map shows the regional location.

peaks fields of 50–150 mT. One sample from each site was further stepwise thermally treated to a maximum temperature of 670°C, and all remaining samples were finally treated at two or three selected cleaning temperatures. This thorough two-stage treatment of all samples [17] is most likely to achieve a complete recovery of the magnetisation record held in igneous rock samples.

Sample and site behaviour with this treatment can be categorised under several headings (as a rule no component with a coercivity less than 10 mT is considered here):

3.1. Group (a): Gabbro-anorthosite remagnetised by post-Jotnian dolerites

Some sites (4, 6, 11 and 16) lying close to the western margin of the post-Jotnian dolerite outcrop (Fig. 2) and all sites to the east lying below the extrapolated extension of this thick sheet-like body (1, 2, 3, 9 and 10) possess a very stable NE magnetisation with negative inclination (Fig. 3): high coercivity remanence exhibits a smooth exponential fall in moment and little directional change with progressive treatment. Thermal demagnetisation of the residual moment shows no further systematic change in direction with a sharp fall in intensity close to the Curie point of magnetite (Fig. 3).

Four further sites (5, 7, 8 and 13) exhibit comparable behaviour to a.f. demagnetisation, but with

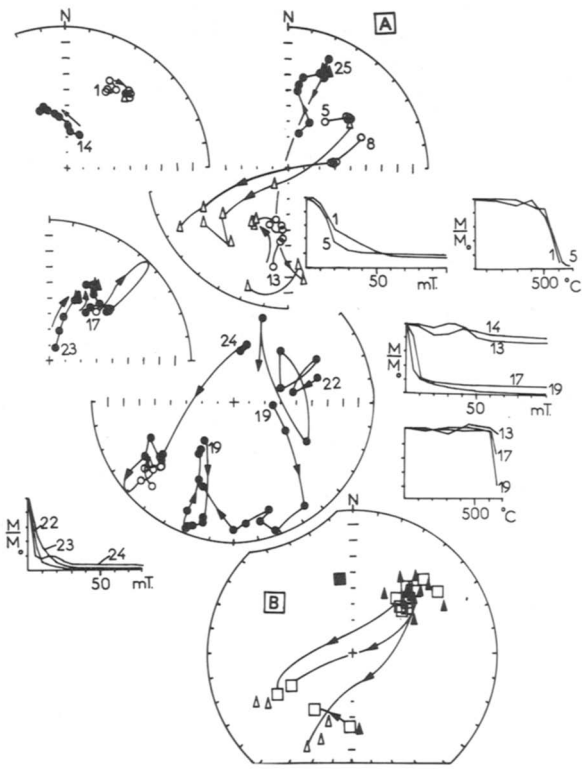


Fig. 3. A. Behaviour of some representative samples from the Nordingrå gabbro-anorthosite with progressive a.f. (circles) and thermal (triangles) demagnetisation; stereographic projections. Positive inclinations are indicated by closed symbols and negative inclinations by open symbols; and magnetic moments are expressed as a fraction of the initial moment M_0. B. The distribution of site mean directions of magnetisation after cleaning of the sites from the anorthositic facies (triangles) and gabbro facies (squares); where more than one medium or high coercivity component has been identified both means are plotted and the trend of increasing coercivity is indicated by the arrowed lines.

TABLE 1

The Nordingrå gabbro-anorthosite complex (18.6°E, 62.9°N) site mean palaeomagnetic statistics after a.f. and thermal treatment

Site No.	N	D	I	R	K	M_0 ($\times 10^{-5}$ A m^2/kg)	Curie point(s) ± 10°C and J_S-T curve type
(1) Anorthositic facies							
1	6	43.1	−26.1	5.8	27	6.00–1.29	565, B
2	6	37.9	−46.4	5.9	63	0.61–0.32	565, D
3	6	50.7	−46.4	5.9	90	0.89–0.13	320, 565, D
4	7	43.3	−48.1	6.9	114	1.03–0.15	335, 580, A
5	(a) 6	44.4	−40.1	5.9	104	0.30–0.06	330, 575, B
	(b) 6	240.5	−47.0	5.7	21		
6	6	53.0	−24.5	5.9	42	0.21–0.05	−/535, D
7	(a) 6	44.4	−34.8	5.9	35	0.22–0.06	575, B
	(b) 4	43.3	40.2	3.4	5		
8	(a) 6	46.4	−42.8	5.9	295	1.30–0.95	335, 565, A
	(b) 6	240.1	−37.2	5.4	9		
9	6	47.7	−48.1	5.9	53	3.57–0.38	575, A
10	6	45.0	−43.5	5.9	74	1.48–0.41	575, A
11	7	38.7	−37.0	6.9	116	2.93–0.03	310, −/530, D
12	6	43.3	−35.4	5.9	98	1.08–0.01	575, A
13	(a) * 6	182.9	−44.4	5.9	51	0.03–0.01	300, 565, D
	(b) 6	214.9	−47.4	5.5	9		
14 *	6	349.6	42.9	5.7	19	0.23–0.03	325, −/545, D
15	6	239.3	−36.6	5.4	9	0.42–0.01	335, −/515, D
(2) Gabbroic facies							
16	6	47.6	−37.9	5.9	187	1.38–0.69	340, 580, D
17	7	39.1	29.4	6.6	13	0.96–0.21	575, A
18	6	47.8	26.8	5.5	11	0.37–0.11	335, 575, B
19 *	6	175.4	42.8	5.4	8	0.93–0.16	555, B
20	3	242.6	−23.3	2.9	43	1.82–0.12	325, 555, A
21	(a) 6	45.1	40.8	5.8	31	0.79–0.01	340, −/575, C
	(b) 5	205.8	−24.9	4.8	22		
22	6	47.8	32.9	5.7	16	1.14–0.61	335, 565, 650, A
23	6	27.9	49.0	5.8	29	0.44–0.21	320, 575, 650, A
24	7	234.9	−20.9	6.1	6	1.10–0.13	310, 555, 620, C
25 *	5	199.1	−33.1	4.8	20	0.83–0.12	335, 575, C
26	7	46.7	41.9	6.3	9	0.03–0.01	−/550, D
27	5	40.5	39.7	4.9	28	0.80–0.29	565, A
28	6	40.1	38.8	5.9	36	0.54–0.08	350, 570, C
29 *	5	197.9	−44.9	4.8	27		300, 570, 650, C
30	4	61.6	46.2	3.8	17	0.29–0.05	345, 570, 660, C
31	6	30.7	34.7	5.9	40	0.98–0.31	355, 570, C
32 **	6	63.1	26.1	5.3	7	0.62–0.09	335, 575, 670, C

Group mean cleaned directions

	N	D	I	R	K	α_{95}	Palaeomagnetic pole
(1) *Nordingrå gabbro-anorthosite, Jotnian remagnetisations:*							
	13 (sites)	45.1	−39.4	12.9	93	4.3°	157.4°E, 2.3°S (dp = 3.1°, dm = 5.2°)
(2) *Nordingrå gabbro-anorthosite, primary magnetisation:*							
	18 (sites)	48.0	37.1	17.6	44	5.3°	138.1°E, 36.9°N (dp = 3.6°, dm = 6.2°)

thermal treatment above the Curie point of magnetite a small residual component is detectable which exhibits marked directional change to achieve NE positive or SW negative directions before the Curie point of hematite is reached; this component generally resides in a few percent of the initial moment (Fig. 3).

The mean of all 13 sites with NE negative directions as the initial (magnetite), or the only detectable, component is $D = 45.1°$, $I = -39.4°$ ($\alpha_{95} = 4.3°$). This is not significantly different from the mean of $D = 47.8°$, $I = -45.5°$ ($\alpha_{95} = 2.6°$) derived from extensive study of the post-Jotnian dolerites in this and adjoining areas [18]. It appears that this large dolerite body has achieved extensive remagnetisation of the adjoining gabbro-anorthosite: close to the upper margin and more extensively below the lower margin (Fig. 2) the thermal effect of the dolerite has been sufficient to completely reset the magnetisations; further from the margin it is possible to recover a primary component residing in hematite.

3.2. Group (b): Gabbro-anorthosites with pre-Jotnian dolerite magnetisations

Further away from the outcrop of the post-Jotnian dolerites, the gabbro-anorthosite facies have a magnetisation unrelated to the dolerites. Some samples possess a large viscous remanence related to the present field direction while a few have a comparable component in a direction similar to the dolerite which can be completely removed by a.f. cleaning (17 in Fig. 3). Stability to a.f. cleaning is variable but is generally less than group (a) with a lower range of median destructive fields (10–30 mT compared with 15–30 mT); site 22 (Fig. 3) is typical of the least stable low coercivity samples. Most of these sites achieve NE positive or SW negative directions also stable to subsequent thermal treatment. There are, however, a few exceptions such as 14 (Fig. 3) which yields an intermediate direction and site 19 (also illustrated in Fig. 3) where separate samples from the same location show different and complex behaviour. Samples either show little or no fall in moment with progressive a.f. treatment, or a dramatic fall followed by no change. With thermal treatment these residual moments change little up to the Curie point of hematite. There are thus at least two components to the remanence of the rocks: one has remanent coercivities in the range 10–50 mT and the other has remanent coercivities in excess of 160 mT. Two polarities are defined by the a.f. cleaning, and although the axis of magnetisation is well defined by this treatment a more complex record is revealed by the subsequent thermal treatment with additional migration of the remanence vector at some sites (e.g. 13, Fig. 3), and complete reversal of the magnetisation vector in some samples and one site (site 21).

By exhaustive demagnetisation tests of the Harp Lake intrusion of Labrador, Irving et al. [19] also identify a complex history of magnetisation in this large anorogenic body similar to the Nordingrå complex; again there is evidence for dual polarities. At Nordingrå the sites unaffected by the post-Jotnian dolerites have quite different properties from sites adjacent to the dolerites. This shows clearly that the remanence is pre-1250 m.y. in age and was acquired in a field of dual polarities; in addition the distribution of polarities is also related to the primary form of the igneous intrusion (see below). The overall group mean direction of magnetisation for the 18 sites from which pre-post-Jotnian dolerite magnetisations have been recovered is $D = 48.0°$, $I = 37.1°$ ($\alpha_{95} = 5.3°$, Table 1).

Thermomagnetic determinations were made on samples from all sites and details are listed in Table 1 with some representative examples illustrated in Fig. 4. While thermal demagnetisation suggests that

TABLE 1, notes:

* Site mean data excluded from overall group mean calculations.

** Monzonitic facies.

D and I are the mean declination and inclination, respectively, of N samples or sites with resultant vector R, Fisher's precision parameter $k = (N-1)/(N-R)$, and α_{95} is the radius of the 95% confidence oval about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively. M_0 = intensity of magnetisation prior to treatment. Curie points are given on heating/cooling where these are different. Curve types are categorised as: A, heating curve = cooling curve, B, cooling curve initially initially below heating curve; C, cooling curve initially above heating curve and pyrrhotite inflection largely or wholly eliminated; D, large increase in J_S on cooling.

hematite is an important remanence carrier at many sites, it is not detected in most of these determinations probably because the saturation remanence (J_s) of hematite (~0.4 A m^2/kg) is much less than that of magnetite (~92.7 A m^2/kg) and it escapes detection unless it forms a major fraction of the magnetic mineral content. A prominent feature on most J_s-temperature curves from the gabbro-anorthosite is a marked Curie point between 300 and 400°C; the heating process clearly promotes a chemical change to this component because this inflection point is generally eliminated on cooling (Fig. 4). In addition to this component most sites exhibit a Curie point in the range 560–580°C attributable to titanomagnetite close to pure magnetite in composition; at five sites, however (Table 1), this Curie point only appears when the samples are cooled.

Microscopic examination of polished sections of the gabbro anorthosite under oil immersion reveals a combination of possible remanence carriers. All sections contain numerous fine rods dispersed and

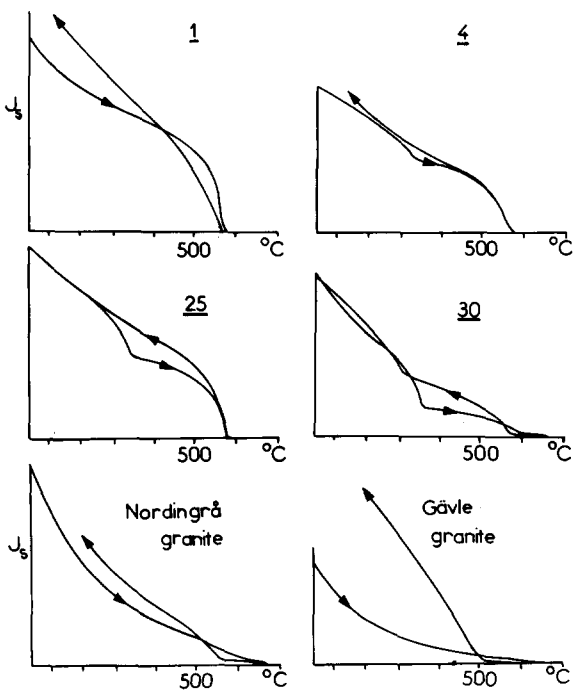


Fig. 4. Some representative thermomagnetic determinations (saturation magnetisation, J_s, versus temperature in °C) from the palaeomagnetic collection; full details for all sites are given in Table 1.

oriented within feldspar megacrysts; these are typically about 1 μm in width (but go down to the limits of resolution), and 5–20 μm in length although a few examples reach 100 μm in length. With few exceptions, these rods are birefringent, and although their small size precludes positive identification, it is probable that they belong to the ilmeno-hematite series. A similar phase has been identified in the Harp and Morin anorthosite bodies of Canada [19,20].

Magnetite occurs most typically as large (mostly 0.1–1.0 mm) discrete and homogeneous grains generally devoid of exsolution and marginal alteration; this mineral is also frequently present as rods, but in contrast to the ilmeno-hematite rods, these are stubby in shape and up to 5 μm in width. A third prominent constituent in most sections is pyrrhotite which must account for the low (300–350°C) Curie point (Table 1) [21,22], and this phase is oxidised to a greater or lesser degree by heating in air (Fig. 4). The pyrrhotite and hematite in these sections can clearly not be cogenetic minerals, and their presence in close proximity appears to reflect the complex petrogenesis of the gabbro-anorthosite facies [14]. In practice the hematite is an exsolution phase within accumulative feldspar, and the large magnetite grains shows clear evidence of being strongly resorbed: where associated with pyrrhotite they form rounded resorbed and anhedral grains with re-entrants. By contrast, the pyrrhotite grains are sometimes euhedral, form ramifying veins, and are always interstitial; they may overgrow the magnetites and commonly form within re-entrants.

Large magnetite grains are usually associated with magnetic remanence of low stability [23], and it is with this phase that the medium coercivity remanence of the gabbro-anorthosite probably resides. By contrast fine needle-shaped grains of both magnetite [24] and hematite [25] are associated with high coercivity remanence with high stability. The fine exsolved rods (apparently predominantly of hematite) within the feldspar megacrysts appear to be the most likely source of high coercivity remanence in the gabbro anorthosite facies of the complex.

When the spatial distribution of the site mean directions is examined (Fig. 2), it is clear that there is a central zone within the gabbro anorthosite outcrop comprising sites 7, 8, 13–15, 20, 21, 24, 25 and 29 which exhibit polarities (equivalent to SE direc-

tions of magnetisation) which are opposite to the bulk of the remaining sites (NE positive directions of magnetisation). They form a broad zone concentric with the outcrop of the later granite (Fig. 2) which would have been parallel to the probable regional isotherms during cooling. One possible explanation for this zone is, therefore, that it cooled through the critical range of blocking temperatures during an interval of opposite polarity to the remainder of the body. Owing to the very large and poorly defined a.p.w. movements in late Precambrian times [26–28], the polarity of the Precambrian field is still uncertain and it is not yet clear whether this represents a normal or reversed field direction.

3.3. Group (c): The Nordingrå granite

Samples were drilled from six locations (Fig. 2) near the central part of the Nordingrå granite outcrop along the fjord Morrfjärded. They possess weak total NRM's (1.63–0.01 A m^2/gm) which exhibit rapid initial fall to 20% or less with progressive a.f. treatment (Fig. 5); directional movement is slight and samples generally retain steep positive directions of magnetisation probably residing in magnetite (Fig. 4). With progressive thermal treatment directions move towards the NE positive and SW negative quadrants and define two approximately anti-parallel groups of high blocking temperature directions (Fig. 5B, since magnetic moments sometimes fall close to the limits of detection at this stage of treatment, the stable end points are not in all cases well established). The thermally cleaned group mean direction of magnetisation for the granite assuming dual polarity is

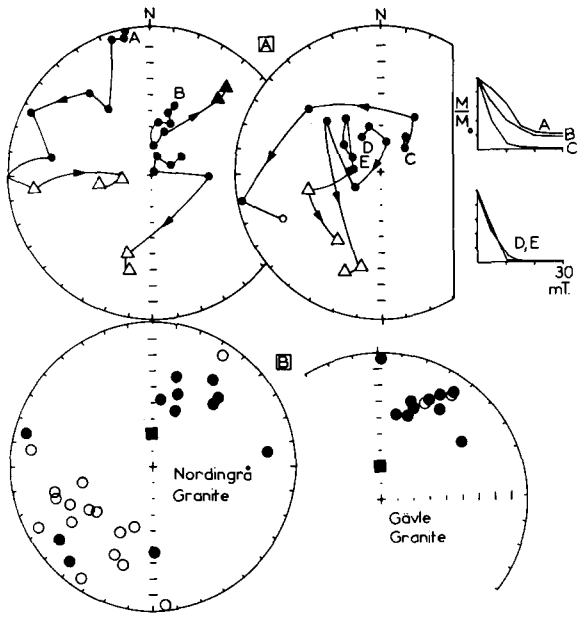


Fig. 5. A. Behaviour of directions of magnetisation with progressive a.f. (circles) and thermal (triangles) cleaning for samples of the Nordingrå and Gävle granites; symbols are as for Fig. 3. B. Distribution of site mean directions of magnetisation of the Nordingrå and Gävle granites after a.f. and thermal cleaning; the square is the direction of the present mean geomagnetic field in the sampling area.

$D = 221°$, $I = -25°$, 28 samples (Table 2); this component probably resides in hematite. The granite high blocking temperature component has slightly shallower inclination than the gabbro-anorthosite, and the granite records a greater percentage of SW negative directions.

TABLE 2

Palaeomagnetic results from the Gävle and Nordingrå rapakivi-type granites of east Sweden, thermally cleaned

N (samples)	*R*	*D*	*I*	*K*	α_{95}	Pole position
Gävle granite						
12	11.0	25.9	17.3	11	13.4	165°E, 35°N (dp = 7.2°, dm = 13.8°)
Nordingrå granite (1415 ± 30 m.y.)						
28	22.8	221.3	−25.1	5	13.3	−31°E, −32½°N (dp = 7.7°, dm 14.3°)

3.4. Group (d): The Gävle granite

The Gävle granite is a smaller rapakivi-type intrusion than the Nordingrå complex with a poorly exposed outcrop at ca. 16.7°E, 60.7°N. Palaeomagnetic results from post-Jotnian dolerites cutting sandstones of possible Jotnian age associated with this body were reported by Poorter [29]. Palaeomagnetic samples from the granite were collected for the present study from quarry exposures between Storvik and Hammerby. With progressive cleaning, these samples move from steep to shallow NE directions of magnetisation and define a weak high blocking temperature component with a mean $D = 26°$, $I = 17°$, ($\alpha_{95} = 13°$); this probably resides in hematite (Fig. 4).

4. Proterozoic apparent polar wander relative to the Baltic Shield

The Precambrian palaeomagnetic data from the Baltic Shield fall conveniently intro three categories: firstly, the oldest body of data is derived from rocks associated with the tectonism of the Svecofennian-Svecokarelian mobile zone and magnetised during stabilisation and uplift of this terrain; these poles are probably mostly older than 1600 m.y. although the (Svecofennian) Loftahammar gabbro [29] was apparently magnetised during uplift between 1660 and 1470 m.y. Secondly, a body of mid-Proterozoic palaeomagnetic data derived from anorogenic intrusions and volcanic/sedimentary suites formed after stabilisation and uplift of the Svecofennian-Svecokarelian terrain; these data include the results of this paper, certain results from Finland [30,31] and western Sweden [32,33]. Although these magnetisations are mostly younger than 1450 m.y., there is probably some overlap with the first group; the Dala volcanics and associated intrusives of western Sweden [32] for example have an overall Rb-Sr isochron age of 1634 ± 42 m.y. [6]. The younger limit to this second group of data is conveniently defined by the intrusion of the post-Jotnian dolerites which are widely observed throughout the Baltic Shield cutting the Svecofennian-Svecokarelian and rapakivi-type rocks. Palaeomagnetic studies of these dolerites in the Dalarna, Nordingrå, Gnarp and Gävle regions of Sweden [18,29,32,33] and the Vaasa, Märket and Satakunta regions of Finland [34–36] identify similar palaeomagnetic pole positions and one uniform polarity [18]. Extensive Rb-Sr studies by Patchett et al. [7] also confirm the unity of this igneous province: 13 of the 19 ages (Rb-Sr, U-Pb and K-Ar isochron) relating to these events lie within the age range 1270–1213 m.y., and it is possible that the Jotnian magmatism was confined within a shorter part of this interval.

The third and youngest body of data derives from rocks intruded, or uplifted and cooled, after the post-Jotnian dolerites, and comes exclusively from the Sveconorwegian mobile zone [25,32,37–39] and basic intrusions bordering this zone [40]. These data and their implications are reviewed by Patchett and Bylund [40] and subsequent discussion here is largely restricted to the second data set.

The palaeomagnetic poles assigned to this mid-Proterozoic interval are listed, with relevant statistical and age data, in Table 3 and plotted in the form of an a.p.w. path in Fig. 6A. In addition to the documentation of these data in the literature (see references), two additional notes are relevant here:

(1) An age of about 1300 m.y. has generally been accepted for the Jotnian (Satakunta) sandstones of Finland (pole 4) [31]. Poorter [29] quotes an age of 1470–1420 m.y. for these rocks although the origin of this assignment is unclear. Lundqvist [41] summarises age data for the major Jotnian outcrop in the Dalarna region of western Sweden (Fig. 1). This postdates the Dala igneous activity (1700–1600 m.y.) but K-Ar ages of 1310–850 m.y. [3] reflect later tectonic/magmatic events. The Jotnian sediments from the Baltic Sea 40 km north of Gotland yield K-Ar ages on authigenic material of 1550 and 1500 m.y. [42]. There is thus some uncertainty about the correct position of poles 3 and 4 in the sequence of Table 3 and Fig. 6A.

(2) Noting that post-Jotnian dolerites from the Dalarna district and the Upper Dala volcanics yielded palaeomagnetic poles close to Palaeozoic rocks from Scandinavia, Priem et al. [6] and Mulder [32] suggested that the magnetisation was overprinted during the Caledonian orogeny. However, Neuvonen [30] found comparable magnetisations in Finnish rocks remote from the Caledonian Front and rejected the remagnetisation hypothesis. Also, Poorter [29] and

TABLE 3

Mid-Proterozoic (ca. 1500–1250 m.y.) palaeomagnetic poles from the Baltic Shield

Pole No.	Rock suite	Age of magnetisation	Sites/ samples	Pole position			Polarity	Reference
				°E,	°N	*dp/dm*		
1	Upper Dala volcanics, Sweden	(>3) <1634	2/11	176,	23	–/–	2	Mulder, 1971 [32]
2	Loftahammar gabbro, Sweden	1660–1470	5/26	179,	23	2/5	2	Poorter, 1976 [29]
3	Jotnian (Dalarna) (a) sediments	(<1, >14)	6/27	177,	33	7/14	m	Piper and Smith, 1979 [33]
	(b) lavas	(<3(a))	8/46	189,	27	5/11	m	Piper and Smith, 1979 [33]
4	Jotnian (Satakunta) sediments, Finland	1470–1420	6/10	180,	3	8/12	2	Neuvonen, 1973 [31]
5	Gävle granite	>1250	2/12	165,	35	7/14	2	Table 2
6	Nordingrå gabbro-anorthosite	⩾1385	18/91	138,	37	4/6	m	Table 1
7	Nordingrå granite	1385	6/28	149.	33	8/14	m	Table 2
8	Ragunda complex	1293	19/113	165,	54	6/9	m	Piper, 1979 [10]
9	Nordingrå complex, Jotnian remagnetisation	1254	13/80	157,	–2	4/6	2	Table 1
10	Vaasa dolerites, Finland	1270–1225	14/15	164,	7	3/6	2	Neuvonen, 1966 [34]
11	Satakunta dolerites, Finland	1236–1225	15/18	158,	2	3/5	2	Neuvonen, 1965 [36]
12	Märket dolerites, Finland	1270	7/9	146,	–6	7/11	2	Neuvonen and Grundström, 1969 [35]
13	Post-Jotnian dolerites, East Sweden *	1254	48/322	157,	–7	2/3	2	Piper, 1978 [18]
14	Post-Jotnian dolerites, West Sweden	(<3) 1215	5/27	159,	5	3/5	2	Piper and Smith, 1979 [33]
			–	149,	–1 **	–	2	Patchett et al., 1978 [7]
				147,	3 **			Patchett et al., 1978 [7]
15	Post-Ragunda dykes, Sweden	(<8) <1293	12/64	184,	15	8/15	m	Piper, 1979 [10]

In polarity column: m = mixed polarities, 1 = N to NE directions of magnetisations, 2 = S to SE directions of magnetisation.

* Overall mean direction calculated from several separate studies listed in Piper [18].

** Virtual geomagnetic poles.

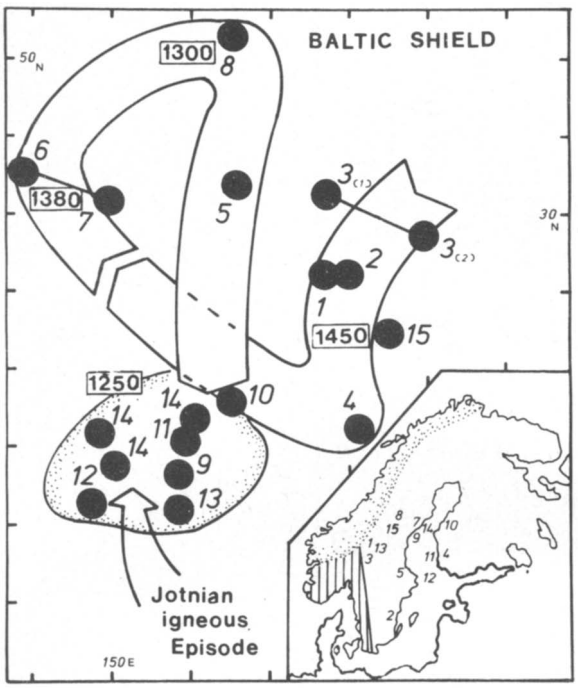


Fig. 6. The a.p.w. path for the Baltic Shield for the interval ca. 1450–1200 m.y. The palaeomagnetic poles are numbered as the data are listed in Table 3 and the inset diagram shows the areal distribution of the palaeomagnetic studies; also indicated are the later Sveconorwegian mobile belt (vertical shading) and the Caledonian orogenic belt.

others [18] resolved magnetisations from the post-Jotnian dolerites of eastern Sweden similar to those from comparable dolerites adjacent to the Caledonian Front. In addition, Spall [43] recognised a polar trend from the Scandinavian data which passed through and beyond the Palaeozoic poles and likewise rejected remagnetisation. Finally, further studies of the Jotnian sandstones and lavas in western Sweden close to the Caledonian Front [33] has identified a remanence quite distinct from the post-Jotnian dolerites by which they are intruded; it is clear that none of these rock units have had their magnetic components significantly reset by later events. The subsequent analysis assumes that all the palaeomagnetic poles in Table 3 are to be assigned to the ages of igneous/metamorphic cooling or sedimentary diagenesis. If this is the case, the collective Middle-Upper Proterozoic data of Table 3 define a closed loop starting with the 1600–1450 m.y. Jotnian and related poles, passing through the 1385 m.y. Nordingrå poles and the 1293 m.y. Ragunda pole, and terminating in the ca. 1270–1210 m.y. post-Jotnian poles.

5. Palaeomagnetic correlations with the Laurentian Shield (1450–1200 m.y.)

Several workers [26,29,44] have correlated the Precambrian palaeomagnetic records of the Baltic and Laurentian Shields. These workers all suggest a similar reconstruction prior to 1200 m.y. with the Sveconorwegian and Grenville mobile belts (ca. 1100 m.y.) in alignment (although this is no longer true [40] during the interval of uplift and cooling of these belts (ca. 1050–750 m.y.)) and the latter two suggest that the two shields were in juxtaposition. Owing to the paucity of the data these analyses have been based on a matching of the general fields of ca. 2000–1000 m.y. palaeomagnetic poles rather than specific a.p.w. trends. Now that the record is more precisely defined, we may look for a correlation between these trends and thus seek to define the validity of the reconstructions.

The contemporaneous data set from the Laurentian Shield (comprising cratonic North America, Greenland and Lewisian foreland of northwestern Scotland) is given in Table 4 and Fig. 7A, and based on two recent compilations [19,49] to which the reader is referred for additional information.

The a.p.w. path commences with the ca. 1400 m.y. Labrador anorthosite poles, passing through the trends defined by the Harp complex and country rocks and the ca. 1300 m.y. Gardar lavas, and terminating in the ca. 1220–1180 m.y. Mackenzie poles. Several of the studies of the Laurentian Shield (3–5, 11 and 12, 16–18 in Table 4) define time sequences of poles (although the duration of the first two is not clear) while the pole position for ca. 1450 m.y. is well represented by the massive anorthosite plutons from Labrador (Michikamau, Harp, Mistastin). Poles with no precise age assignment but which appear to belong in part to this interval include the Grand Canyon and Belt Supergroups (Fig. 7A). The data define a double loop migrating westwards through approximately 35° of longitude and the analogy with the contemporane-

TABLE 4

1450–1200 m.y. palaeomagnetic poles from the Laurentian Shield

Pole No.	Rock unit	Age	Palaeomagnetic pole		Polarity
			°E,	°N	
1	Croker Island complex	1444	217,	−5	2
2	Sherman granite	1380	209,	−8	m
3	Harp complex	1450	206,	2	m
4	Harp dykes	(<unit 3)	228,	19	2
5	Harp country rocks	–	216,	14	m
6	Michikamau anorthosite	1450–1400	{215,	1	2
			{218,	−2	
7	St. Francois rocks	1371	216,	1	2
8	Mistatin pluton *	1313	202,	−1	m
9	Seal Lake Group	1322	205,	6	m
10	Sibley Group	1341	216,	−21	m
11	Lower Gardar lavas, Greenland	>1283	223,	11	m
12	Upper Gardar lavas, Greenland **	>1283	206,	20	m
13	Muskox pluton	1250	185,	4	2
14	Coppermine Group	1257	183,	1	2
15	Mackenzie diabase	1240	171,	1	2
16	Lamprophyre dykes, Greenland	1249–1228	189,	9	2
17	NNW dolerite dykes, Greenland	(<16) 1250–1212	185,	14	2
18	Kûngnât complex	(<17) 1219	182,	9	2
19	Sudbury dykes	1205–1189	179,	−2	2

Based on the compilations of Irving et al. [19] and Piper [49] with ages recalculated [70]. The Greenlandic poles are rotated according to the reconstruction of Bullard et al. [72].

* Irving et al. [19] argue that this pole is approximately 100 m.y. older than the Rb-Sr whole-rock isochron.

** In common with the Jotnian lavas and sediments (text and Piper and Smith [33]) one thick section of the Gardar lavas exhibits anomalous field directions yielding palaeomagnetic poles not listed here; this unit is intruded by the Motzfeldt Complex (1283 ± 31 m.y.) but is not thought to be greatly older than this.

The polarity column uses the same convention as Table 3, and error limits to poles and ages are given in references 19 and 49.

ous data set from the Baltic Shield is thus very close. Also, both these data sets terminate in a series of poles defined by the extensive episodes of dolerite intrusion included within the Mackenzie (Canadian Shield), Gardar (Greenland) and post-Jotnian (Baltic Shield) igneous provinces; these are all of the same polarity if the poles of Fig. 6 and 7A are considered [18] and the extensive Rb-Sr studies of Patchett et al. [7] demonstrate that they are also broadly contemporaneous.

In comparing the a.p.w. records of Figs. 6 and 7 it is not possible to completely superimpose the post-Jotnian and Mackenzie/Gardar poles and also match the ca. 1450 m.y. data without violating the major criterion of a continental reconstruction, that the sialic shields should not overlap. This may reflect a small real difference in age between the Jotnian and Mackenzie activity [7, table 3]. The reconstruction which does not violate this criterion while satisfying the distribution and trends of the poles from the respective shields is illustrated in Fig. 8 and is equivalent to an anticlockwise rotation of 64° about a Euler pole at 1°E, 36°N.

This reconstruction is closely similar to that derived by earlier workers and achieves a partial overlap of the post-Jotnian and Mackenzie/Gardar poles (Fig. 8A) with a close agreement between the ca. 1450 m.y. poles. The post-1450 m.y. polar trend from the Laurentian data is also closely matched by a contemporaneous trend in the Baltic Shield data (Fig. 8). Finally, the polar trend between Ragunda (1293 m.y.) and the post-Jotnian dolerites (ca. 1240

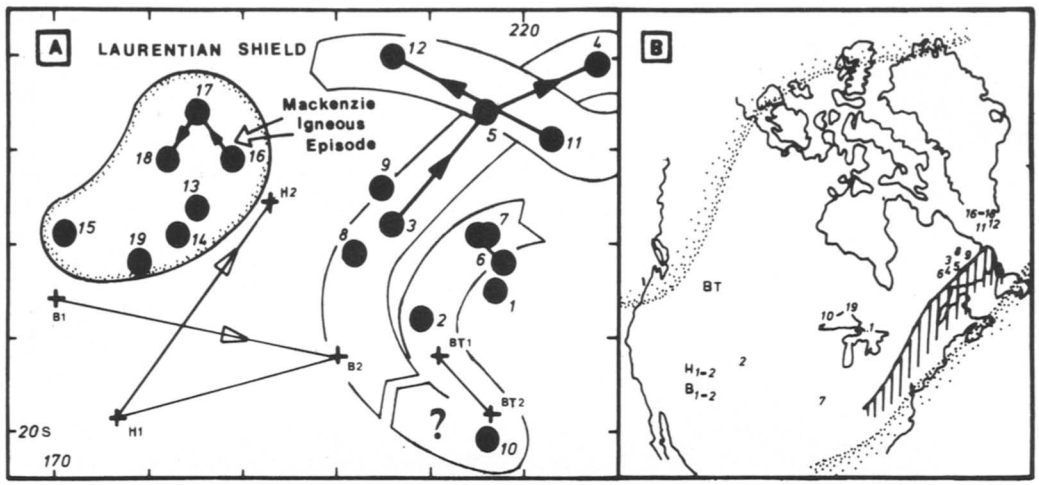


Fig. 7. A. The a.p.w. path for the Laurentian Shield for the interval ca. 1450–1200 m.y.; the stratigraphic sequence of poles is indicated where this is known and poles are numbered according to the listing of the data in Table 4. Palaeomagnetic poles from sedimentary suites probably falling within this general interval are also indicated by crosses and are lettered: BT = Belt Supergroup; B1, 2 = Bass Limestone, and H1, 2 = Hakatai Shades (Grand Canyon Supergroup). B. The areal location of the palaeomagnetic studies of A and Table 4 within the Laurentian Shield; the later Grenville mobile belt is indicated by vertical shading.

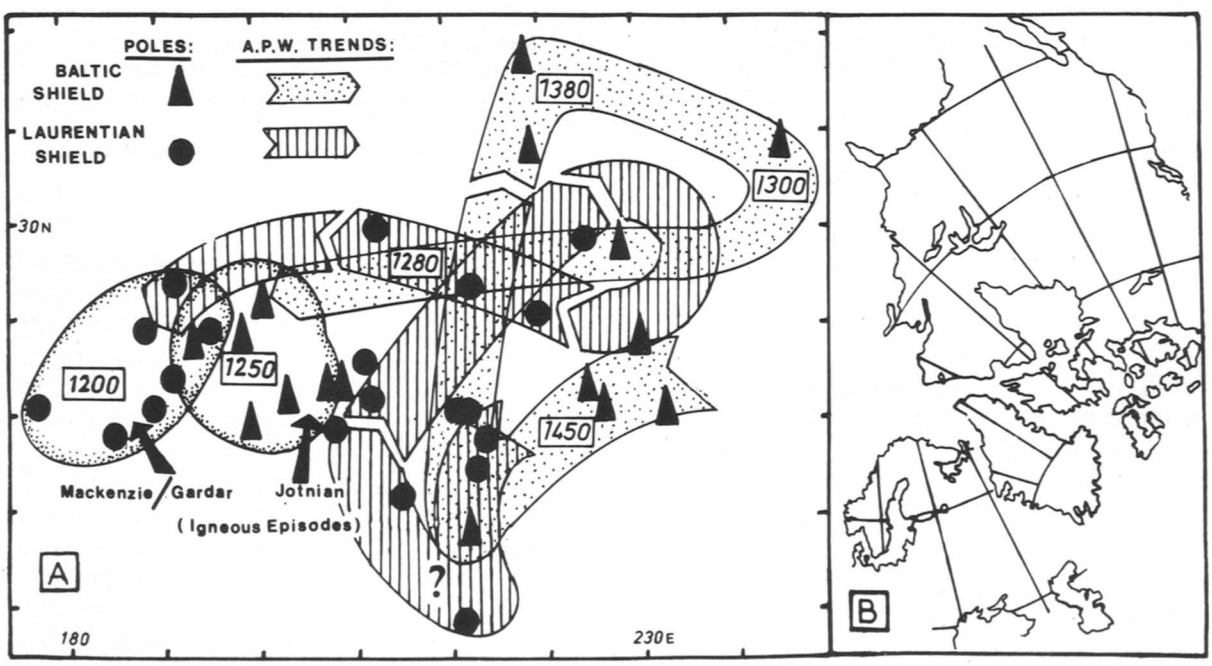


Fig. 8. A. Comparison of the ca. 1450–1200 m.y. a.p.w. paths of the Baltic and Laurentian Shields and the distribution of poles from the two shields. B. The reconstruction of the shields obtained by superimposing the a.p.w. loops from the two shields as indicated in A.

m.y.) is matched by the polar trend identified within the Gardar succession of South Greenland [49] slightly older than the Motzfeldt complex (1283 ± 31 m.y.), and the Gardar dykes contemporaneous with the Mackenzie episode of the Canadian Shield (1250–1212 m.y. [7]). The only discrepancy is Laurentian pole 10 from the Sibley Group [50] with an assigned Rb-Sr age on sedimentary material of 1340 m.y.: the polar swathe of Fig. 8 predicts that this unit may predate the Labrador anorthosites and correlate with pre-1400 m.y. Baltic Shield poles.

There are two ways of testing the validity of the continental reconstruction in Fig. 8: firstly, we may test for correlation between pre-1450 m.y. palaeomagnetic poles from the two shields, and secondly, for correlation between geological features. These are now examined in turn.

6. Correlation of pre-1450 m.y. palaeomagnetic poles between the Baltic and Laurentian Shields

Pre-1450 m.y. poles from the Baltic Shield (Table 5) come predominantly from basic rocks within the Svecokarelian-Svecofennian mobile belts with ages of intrusion in the range of 2000–1900 m.y. [3,5,43, 51]. However, the magnetisations are probably in all cases appreciably younger than this because they date from the uplift and cooling of these mobile zones (ca. 1800–1600 m.y.), and the critical range of blocking temperatures is both lowered and broadened by protracted cooling during uplift [52]. The Tarendo gabbro (pole 1, Table 5) for example was intruded by late-tectonic granites at 1845–1775 m.y. [5,53] and K-Ar uplift ages from the Svecokarelian zones average 1800–1700 m.y. [3]; it is with Laurentian poles of this age that we should correlate the syntectonic gabbro-diorite poles of Table 5.

The Middle-Lower Proterozoic palaeomagnetic data from the Laurentian Shield are summarised and discussed by several workers [46,47,54,55] and poles with assigned ages in the range ca. 1800–1500 m.y. are plotted as circles in Fig. 9. They define part of a large a.p.w. loop and the trend of a.p.w. movements over the interval 1800–1650 m.y. is now well established by the sequence of uplift and cooling magnetisations in the Hudsonian-Nagssugtoqidian-Ketilidian mobile terrains of Greenland and the Canadian Shield [47,54,55]. The youngest part of this loop (1650–1500 m.y.) is currently only defined in outline from the sequence of uplift magnetisations in the Lewisian terrain of northwestern Scotland [55] and amphibolite facies rocks of the Nagssugtoqidian belt of West Greenland.

TABLE 5

Pre-1450 m.y. palaeomagnetic poles from the Baltic Shield

Pole No.	Rock unit	Age	Palaeomagnetic pole		Rotated pole	
			°E,	°N	°E,	°N
1	Tärendö gabbro, Sweden [1]	Svecofennian/ Svecokarelian (see text)	229,	45	276,	−4
2	Ylivieska gabbro, Finland [2]		242,	43	285,	−7
3	Pohjanma gabbro-diorite, Finland [2]		240,	38	282,	−12
4	Tammela gabbro-diorite, Finland [2]		259,	53	296,	3
5	Mikkeli gabbro-diorite, Finland [2]		230,	53	279,	4
6	Hyvinkää gabbro-diorite, Finland [2]		246,	44	288,	−7
7	SW Finland gabbro-diorites [2]		243,	40	285,	−10
8	Åva intrusives, Finland [3]	1690 [6]	169,	41	239,	19
9	Häme dolerites, Finland [4]	1670–1620 [7]	72,	67	290,	62
10	Kumlinge dolerites, Finland [5]	1620 [8]	204,	8	238,	−26
11	Föglö dolerite, Finland [5]	>1620 [8]	187,	31	241,	2

[1] Cornwell, 1968 [53]; [2] Neuvonen, 1974 [51]; [3] Neuvonen, 1970 [30]; [4] Neuvonen, 1967 [74]; [5] Neuvonen and Grundström, 1969 [35]; [6] U-Pb determinations on associated rocks; [7] Age inferred from assocation with Rapakivi intrusions; [8] according to Neuvonen and Grundström, unit 10 is probably somewhat older than rapakivi granite (1620 m.y.) and unit 11 is definitely older.

Using the reconstruction of Fig. 2 the rotated positions of the contemporaneous Baltic Shield palaeopoles are listed in Table 5 and plotted as triangles in Fig. 9. A close correspondence between the Baltic Shield and Laurentian poles is immediately clear: the Svecofennian-Svecokarelian poles 1–7 agree closely with the post-Hudsonian and related poles (1800–1700 m.y.), suggesting a genetic link between these mobile episodes and their uplift/cooling magnetisations already apparent from the age data [3,5,29,51]. Furthermore, the post-tectonic dolerites of Finland (poles 8–11, Table 5) related in part to rapakivi magmatism in that region (1700–1600 m.y.) correlate closely with the younger extension of the Laurentian loop in Fig. 9. This analysis thus reinforces earlier and more general assessments [26,29] suggesting that the Baltic and Laurentian palaeomagnetic data fit a single continental reconstruction prior to 1200 m.y. and implies that the reconstruction extends at least as far back as the ca. 1800 m.y. Hudsonian, Svecofennian and related mobile episodes. Since further refinement of this conclusion will no doubt be possible as more data are forthcoming, the discussion of geological implications that follows is general rather than specific in nature.

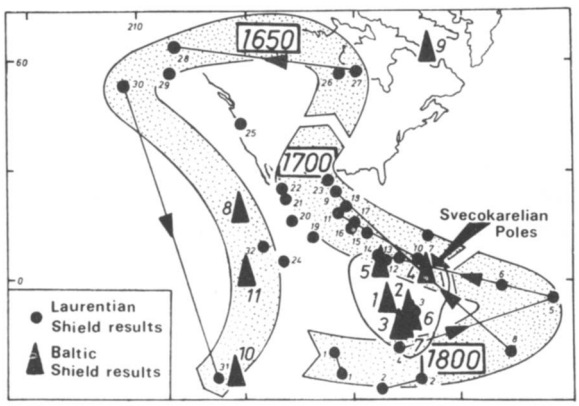


Fig. 9. A comparison of pre-1450 m.y. palaeomagnetic poles (triangles) from the Baltic Shield with contemporaneous Laurentian Shield data (circles) on the reconstruction of Fig. 8; a generalised a.p.w. path from the Laurentian data based on references 47, 54 and 55 is indicated by the stippled swathe. The Laurentian poles with ages in brackets are numbered: 1 = Seward subgroup (1800); 2 = Retty "B" magnetisations; 3 = Martin Formation (1944–1647); 4 = Pearson Formation "A" magnetisations (1800); 5 = Pearson Formation "B" magnetisations; 6 = Et-Then Group (1848–1642); 7 = Sparrow dykes (1700); 8 = Flin-Flon Group secondary magnetisations (1800); 9 = Flin-Flon Group tertiary magnetisations (1700–1600); 10 = Kahocella primary magnetisations (1873); 11 = Kahocella secondary magnetisations; 12 = Pearson Formation "C" magnetisations; 13 = Menihek "B" magnetisations; 14 = Dubawnt Group (1787); 15 = Nonacho sediments; 16 = Kaminak dykes (1905–1627); 17, 18 = Itivdleq dykes (1843–1712); 19 = Daly Bay "A" magnetisations (1634); 20 = Cape Smith metavolcanics (1663–1460); 21 = Kangamiut dykes (1803–1662); 22 = Godhåb gneisses; 23 = Nordre Stromfjord granulites and amphibolites; 24 = Seward "B" magnetisations; 25 = Wind River dykes (1894–1692); 26 = Sagdlerssuaq amphibolites (1662); 27, 28 = Ketilidian metavolcanics (1743–1612); 29 = West Greenland amphibolite boudins; 30 = Lewisian "A" magnetisations (1650–1625); 31 = Lewisian "C" magnetisations; 32 = Western Channel diabase (1792–1410). The age sequence of the poles is indicated by the arrowed lines where this is known and Greenlandic poles are adjusted using the conventional reconstruction [72]. The contemporaneous Baltic Shield poles are listed in Table 5.

7. Correlation of geological features between the Baltic and Laurentian Shields

The Proterozoic reconstruction of Fig. 8 emphasises the close association in space as well as in time [1,5,6] of the ca. 1800 m.y. Svecofennian-Svecokarelian and Hudsonian mobile episodes (Fig. 10). The crustal regions which stabilised in Middle-Lower Proterozoic times exhibit only weak anisotropy, although a strong fabric has been imprinted on the crust within narrow linear zones of strike-slip movement [56]. The trend of these lineaments within the Baltic-Ukrainian Shield becomes parallel, or nearly so, with those in the Laurentian Shield on the continental reconstruction; structural trends within the ca. 1800 m.y. mobile belts of these two shield regions (although variable) are also broadly aligned (Fig. 10), and place in a wider context the observation of several workers (e.g. [57,58]) that Archaean and Proterozoic structural trends are closely parallel to one another over large areas of the shields. Both these strike slip lineaments and the mobile belt structural trends are also parallel to the long axis of the continental unit defined by the palaeomagnetic reconstruction (Fig. 10).

The most notable feature of mid-Proterozoic development of these shields was the intrusion of massive anorthosite and related granite, monzonite, gabbro

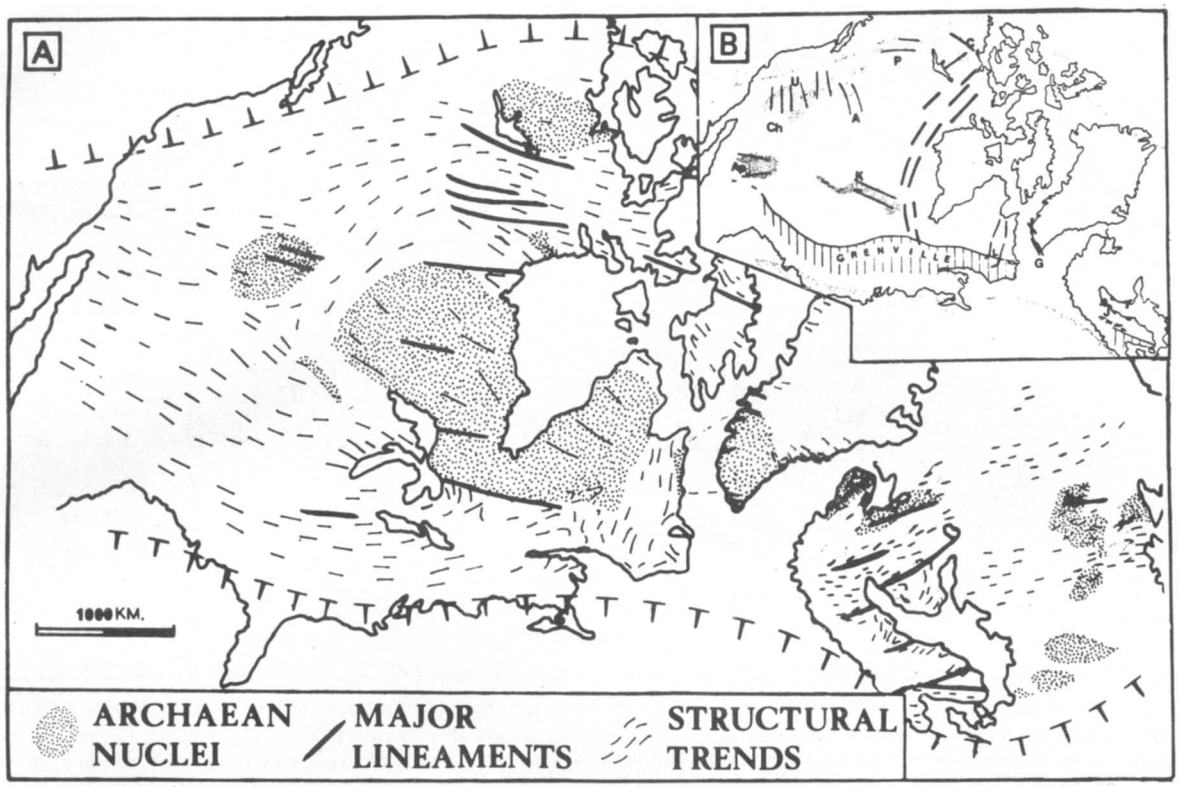


Fig. 10. A. The distribution of Archaean regions, major late Archaean/Lower Proterozoic lineaments [56,57] and generalised Proterozoic structural trends (based mainly on references 1 and 2 and the metallogenic maps of Europe) on the reconstruction of the shields derived from pre-1200 m.y. palaeomagnetic data (Figs. 8 and 9). B. Generalised distribution of the ca. 1200 m.y. Mackenzie-Sudbury-Gardar-Jotnian intrusive activity (thick lines), the ca. 1100 m.y. Grenville Sveconorwegian mobile belts (vertical lines), the distribution of Middle/Upper Proterozoic postulated rifts and anorogenic sedimentary suites (stipples). Named rifts are indicated as follows: A = Apache; Ch = Chuar; A = Alberta; P = Purcell; C = Coronation; K = Keweenawan; L = Labrador; G = Gardar; and J = Jotnian.

and syenite anorogenic plutons which extend from western North America to the Urals [1,57,58]. They are found predominantly within the terrain subjected to widespread mobile activity between 2000 and 1700 m.y. [2], while within the Grenville and Sveconorwegian belts (ca. 1100 m.y.) they appear to just predate the latest stages of tectonism but may in part be reworked older intrusions [1,2,59]. The distribution of these bodies is plotted on the palaeomagnetic reconstruction in Fig. 11 after Bridgwater and Windley [2] and Memoir 18 of the New York State Museum Science Service; alkaline intrusions are also summarised by Barker [60].

A further major characteristic of the Proterozoic igneous activity is the progressive contraction of this anorthosite/rapakivi-type magmatism across the continental reconstruction of Fig. 11. This was first noted within the individual shields by Bridgwater and Windley [2] who related the young mineral ages within this zone of magmatism to high thermal activity. They also observed that magmatism followed closely on the ca. 1800 m.y. mobile activity near the extremities of the shield and terminated early, but near the Grenville and Sveconorwegian belts it continued until the ca. 1100 m.y. tectonism; it is here that the largest time discrepancies are found between geological events and anomalously young mineral ages.

Within the Laurentian Shield this migration of activity is expressed by the contraction of the rapakivi

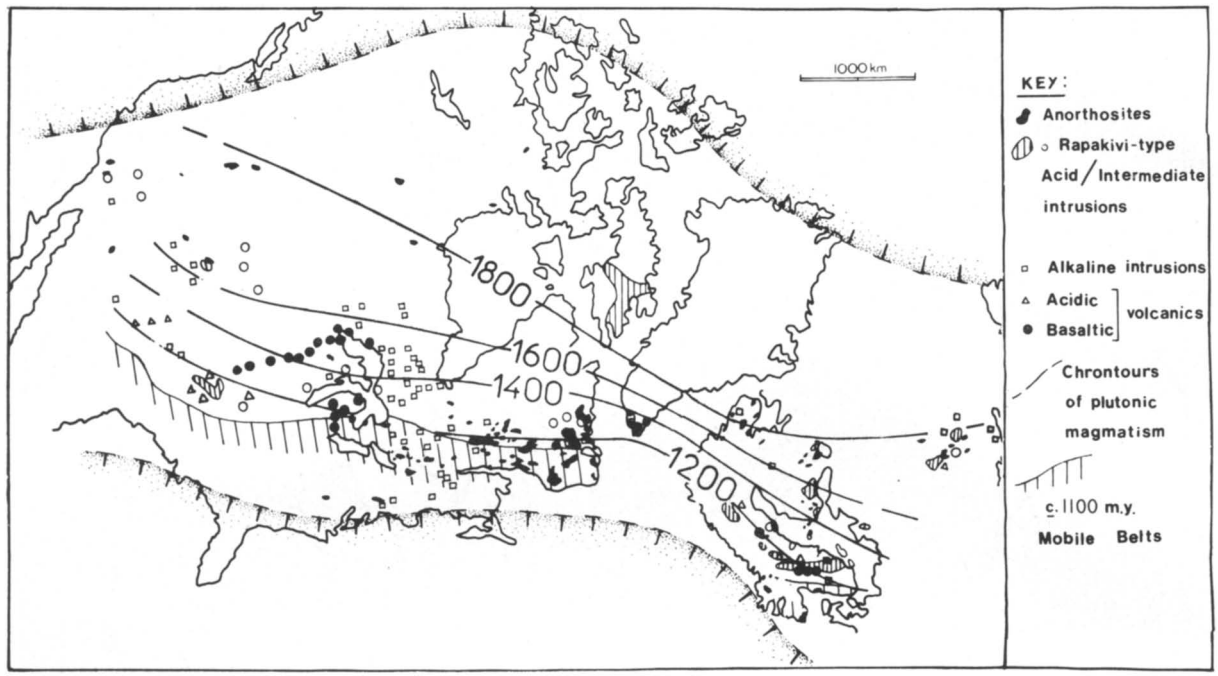


Fig. 11. The distribution of Middle/Upper Proterozoic magmatic-tectonic activity within the Laurentian-Baltic-Ukrainian Shields (based mainly on reference 2). The "chrontours" define the approximate age of *termination* of the characteristic Proterozoic rapakivi-anorthosite-alkaline magmatism; they demonstrate the gradual restriction of this activity towards one active margin of the supercontinent (which is also defined by the ca. 1100 m.y. Grenville and Sveconorwegian mobile belts) as temperature gradients diminished during this epoch.

magmatism of the western U.S.A. between 1600 and 1350 m.y. and alkaline intrusion in the interval 1500–1170 m.y. (e.g. [1,2,61]). Further east, anorthosite magmatism in the Churchill Province ceased shortly after 1800 m.y. while in Labrador the major episodes of intrusion were distributed between 1500 and 1300 m.y. [19] continuing to 1200 m.y. in Minnesota and 1100 m.y. within the Grenville Province [1]. In southwest Greenland the rapakivi intrusion closely followed the Ketilidian mobile episode at 1780 m.y. but continued at depth to 1200 m.y. within the Gardar Province (and close to the extrapolated northern margin of the Grenville Belt).

In the Baltic Shield we again observe the contraction of rapakivi-type magnetism from Finland (ca. 1640 m.y.), the Kola Peninsula (alkali granites and syenites, ca. 1700 m.y.) and north Sweden (1700–1500 m.y., see [62–65]) to central Sweden (1400–1300 m.y., section 2 of this paper) and southern Norway where the Rogaland and related anorthosites are approximately synorogenic with respect to the Sveconorwegian Belt and dated 1000–950 m.y. [37,59]. In Fig. 11 "chrontours" drawn to embrace the times of termination of anorthosite/rapakivi magmatism are again parallel to the leading edge of the continental reconstruction and the Grenville-Sveconorwegian tectonic lineament. They define the gradual contraction of the zones of high heat flow as this major body of crust consolidated and cooled through Proterozoic times, and are unlikely to reflect a significant change in crustal level across the shields [66]. Models linking the range of mid-Proterozoic tectonic and magmatic features [1,67] require high-temperature gradients and partial mobilisation of the crust and possibilities for these igneous suites to develop within an anorogenic environment became progressively more restricted (Fig. 11) towards the active margin ceasing altogether about 1000 m.y.

Three further aspects of the reconstruction of Fig. 8 deserve comment. Firstly, the ca. 1250 m.y. Jotnian, Gardar, Mackenzie, and Sudbury suites of dykes and other minor intrusions define two broad arcuate zones spanning most of the width of the shield (Fig. 10B). Secondly, the mid-Proterozoic grabens (or aulacogens) which may reflect early stages of brittle fracturing and failed attempts at disruption of the continental unit [68] are concentrated near the "passive" leading edge of the reconstruction (Fig. 10B). Thirdly, the Grenville Belt, the Grenville Front, the Sveconorwegian Belt and the "schistosity zone" separating this latter belt from the remainder of the shield are closely aligned on the continental reconstruction (Fig. 11). This links in a spatial context these belts of thermal/tectonic reactivation at ca. 1100 m.y. [40] which both experienced protracted uplift and cooling following this episode [3,69].

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